

Qatar-1b

R-0483

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_200715 · created 2026-07-20T07:10:32+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459045.7547 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.055
Transit depth (Rp/Rs)^2	3.68 +/- 2.13 [%]
Orbital Inclination (inc)	88.88 +/- 2.84
Airmass coefficient 1 (a1)	1.72 +/- 0.25
Airmass coefficient 2 (a2)	-0.38 +/- 0.41
Scatter in the residuals of the lightcurve fit is	15.5 %
Best Comparison Star	#3 - [412, 299]
Optimal Aperture	3.74
Optimal Annulus	11.11
Transit Duration (day)	0.0831 +/- 0.0072

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.192 ± 0.055 vs 0.1463 (deviation 0.83σ)
Duration fit vs published	0.0831 d vs 0.0692 d (deviation 1.94σ)
Mid-time O–C	4.6 min
Depth SNR	3.5
Residual scatter	15.5 %

Light curve

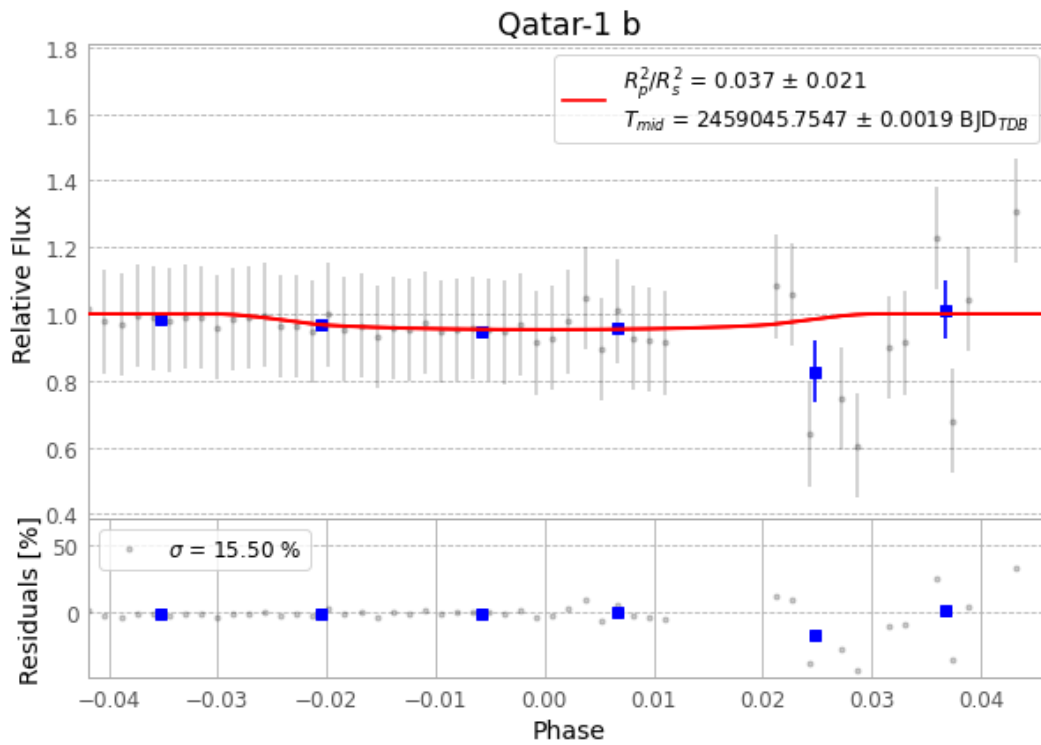


Figure 1 — FinalLightCurve_Qatar-1 b_2020-07-14.png (SHA-256 30ff7e7712b18948...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [483, 278], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 454573d9da56c0b4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

64 FITS frames (42 MB), Qatar-1200715043912.FITS → Qatar-1200715075115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-200715.log (139be0618c24...), FinalLightCurve_Qatar-1 b_2020-07-14.png (30ff7e7712b1...), FinalLightCurve_Qatar-1 b_2020-07-14.csv (431731032626...)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 07:10Z.